

ASHLEY (YE) GAO

🌐 <https://lindagaw.github.io> ✉ ygao18@wm.edu

📍 540 Landrum Dr Room 2381, Williamsburg, VA 23185

EMPLOYMENT

William & Mary <i>Assistant Professor in Computer Science</i>	<i>2023 - Present</i>
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EDUCATION

University of Virginia Advisor: John A. Stankovic <i>Ph.D. in Computer Science</i>	<i>2019 - 2023</i>
University of Virginia <i>M.S. in Computer Science</i>	<i>2017 - 2019</i>
University of California, San Diego <i>B.S. in Computer Science</i> <i>B.A. in Literatures of the World</i>	<i>2012 - 2017</i>

RESEARCH STATEMENT

My research lies at the intersection of **affective computing**, **audio-language modeling**, and **robust machine learning**. I develop methods for **emotion-aware representation learning and generation**, with a focus on audio large language models, controllable text-to-speech, and speech emotion recognition. My work combines advances in **representation learning**, **generative modeling** with applications in **computational healthcare** and **human-centered AI**.

SERVICES

Editorial Roles S5. Associate Editor. Elsevier Smart Health.	<i>2024 - Present</i>
Grant Review Panels (Invited)	
S4. Panelist. National Institutes of Health (NIH).	<i>2025</i>
S3. Panelist. National Institutes of Health (NIH).	<i>2024</i>
S2. Panelist. National Science Foundation (NSF).	<i>2024</i>
Conference Leadership	
S1. Demo & Poster Chair. IEEE/ACM CHASE (Connected Health: Applications, Systems and Engineering Technologies).	<i>2024</i>

PROFESSIONAL MEMBERSHIPS

P2. Association for Computing Machinery (ACM). Member.	<i>2025 - Present</i>
P1. Institute of Electrical and Electronics Engineers (IEEE). Member.	<i>2025</i>

SELECTED PUBLICATIONS

Conference and Journal Publications

[ICML'26] Du, H., Shi, J., Lu, S., Zhou, G., **Gao, Y.**, "Sparse Autoencoders for Interpretable Emotion Control in Text-to-Speech." Proceedings of the International Conference on Machine Learning (ICML), 2026.

- [**ACL'26**] Nair, T., Nafee, M. W., Jiang, M., **Gao, Y.**, Chen, H., Zhang, Y., “Confidence-Aware Ranker Ensembles for Robust In-Context Knowledge Editing.” Findings of the Association for Computational Linguistics (ACL), 2026.
- [**ACL'26**] Shi, J., Du, H., Song, X., Hong, Y. A., Zhang, Y., **Gao, Y.**, “AffectCodec: Emotion-Preserving Neural Speech Codec for Expressive Speech Modeling.” Findings of the Association for Computational Linguistics (ACL), 2026.
- [**ICASSP'26**] Shi, J., Du, H., He, Y., Hong, Y. A., **Gao, Y.**, “Emotion-Aligned Generation in Diffusion Text-to-Speech Models via Preference-Guided Optimization.” IEEE ICASSP, 2026.
- [**ICASSP'26**] Shi, J., Du, H., Hong, Y. A., **Gao, Y.**, “EMO-TTA: Improving Test-Time Adaptation of Audio-Language Models for Speech Emotion Recognition.” IEEE ICASSP, 2026.
- [**ICASSP'26**] Shi, J., Du, H., Hong, Y. A., **Gao, Y.**, “Plug-and-Play Emotion Graphs for Compositional Prompting in Zero-Shot Speech Emotion Recognition.” IEEE ICASSP, 2026.
- [**ICASSP'26**] Mersa, O., Jiang, M., **Gao, Y.**, Li, Q., Zhang, Y., “Hierarchical Convolution Multibranch Transformer for EEG Signals.” IEEE ICASSP, 2026.
- [**ICASSP'26**] Jiang, M., Chen, Y., Liu, G., Sahu, A., **Gao, Y.**, Zhang, Y., “Pairs, Not Labels: Predicting Protein-Phenotype Associations via Link Prediction.” IEEE ICASSP, 2026.
- [**EMNLP'25**] Du, H., Shi, J., Myerston, J., Lu, S., Zhou, G., **Gao, Y.**, “Role-Guided Annotation and Prototype-Aligned Representation Learning for Historical Literature Sentiment Classification.” Findings of EMNLP, 2025.
- [**InterSpeech'25**] Du, H., Lu, S., Zhou, G., **Gao, Y.**, “Emotion-Aware Audio Large Language Models with Dual Cross-Attention and Context-Aware Instruction Tuning.” InterSpeech, 2025.
- [**InterSpeech'25**] Shi, J., Zhang, Y., **Gao, Y.**, “Contrastive Learning for Speech Emotion Domain Generalization via Soft Prompt Tuning.” InterSpeech, 2025.
- [**CHASE'25**] Clapham, J., Zhou, M., MacDonald, C., Koltermann, K., **Gao, Y.**, Shao, H., “ElectroMeter: The Practical Electrolyte Measurement System.” IEEE/ACM CHASE, 2025.
- [**JMIR'25**] Ko, E., **Gao, Y.**, Wang, P., Wijayasingha, L., Wright, K. D., Gordon, K. C., Wang, H., Stankovic, J. A., & Rose, K. M. “Challenges to recruiting dementia caregiving dyads in community-based settings.” JMIR Formative Research, 2025.
- [**TMLR'24**] Macdonald, C., Chu, Z., Stankovic, J., Zhou, G., Shao, H., **Gao, Y.**, “MiddleGAN: Generating Inter-domain Images with GANs.” Transactions on Machine Learning Research, 2024.
- [**CHASE'24**] Koltermann, K., Clapham, J., Blackwell, G., Jung, W., Burnet, E., **Gao, Y.**, Shao, H., et al., “Gait-Guard: Turn-aware Freezing of Gait Detection for Non-intrusive Intervention Systems.” IEEE/ACM CHASE, 2024. (**Best Paper Award**)
- [**SmartComp'23**] **Gao, Y.**, Baucom, B., Gordon, K., Rose, K., Wang, H., Stankovic, J. “E-ADDA: Unsupervised Adversarial Domain Adaptation Enhanced by a New Mahalanobis Distance Loss.” IEEE Smart-Comp, 2023.
- [**HEALTH'21**] **Gao, Y.**, Salekin, A., Gordon, K., Rose, K., Wang, H., Stankovic, J. “Emotion Recognition Robust to Indoor Environmental Distortions Using Out-of-Distribution Detection.” ACM Transactions on Computing for Healthcare, 2021.
- [**PERCOM'21**] **Gao, Y.**, et al. “Out-of-the-Box Deployment to Support Research on In-Home Care of Alzheimer’s Patients.” IEEE Pervasive Computing, 2021.

[JAN'21] Rose, K. M., et al., **Gao, Y.**, "Smarthealth technology study protocol for dementia caregiving relationships." Journal of Advanced Nursing, 2021.

[SENSYS'20] **Gao, Y.**, et al. "A Monitoring and Recommendation System for In-Home Caregivers (Demo)." ACM SenSys, 2020.

Full publication list: <https://scholar.google.com/citations?user=XvbRi5wAAAAJ&hl=en>

RESEARCH LAB

Mentoring Ph.D. Students

M3. Jiacheng Shi, Ph.D. Student, William & Mary.

First author papers: ICASSP, ACL, Interspeech 2024 - Present

M2. Hongfei Du, Ph.D. Student, William & Mary.

First author papers: ICML, ACL, EMNLP, Interspeech 2024 - Present

Undergraduate Students

M1. Lilly Liu, Undergraduate Student, William & Mary.

Initial placement: Carnegie Mellon University (CMU). 2024 - Present

Selected Research Directions

[D3.] Domain-Generalized Audio Sensing — robust speech emotion recognition under distribution shift

[D2.] Sentiment Steering in LLMs — controllable generation of emotional tone in language models

[D1.] Audio Stress Detection for ADRD Caregivers — mobile and scalable systems for caregiver support ([Linda Gaw][1])

HONORS AND AWARDS

Research Awards

H7. Best Paper Award, IEEE/ACM CHASE. 2024

H6. Best Paper Runner-Up Award, IEEE SmartComp. 2023

H5. Best Poster Award, Computer Science Symposium, University of Virginia. 2022

Service Awards

H4. Best Service Award, IEEE/ACM CHASE. 2024

Fellowships and Academic Honors

H3. Ph.D. Fellowship, Computer Science, University of Virginia. 2019

H2. Academic Excellence Fellowship, Computer Science, University of Virginia. 2017

H1. Honors with High Distinction, University of California, San Diego. 2016

REVIEWER SERVICE

S8. Primary Reviewer. ICML, AAAI, ICLR. 2025

S7. Primary Reviewer. NeurIPS, AAAI, ICLR, ACM Health. 2024

S6. Primary Reviewer. AAAI, NeurIPS, ICLR. 2023

S5. Primary Reviewer. IEEE Transactions on Affective Computing. 2022

S4. Primary Reviewer. AAAI. 2022

S3. Primary Reviewer. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies. IMWUT/UbiComp. 2022

S2. Primary Reviewer. 36th Conference on Neural Information Processing Systems. NeurIPS. 2022

S1. Primary Reviewer. ACM Transactions on Computing for Healthcare. HEALTH. 2021

PUBLISHED CREATIVE WRITING

Forthcoming Books

C3. *The Tempest's Call*. Pegasus Publishers, United Kingdom. (Forthcoming) 2026

C2. *The Celestial Blade*. Pink Hydra Press, South Africa. (Forthcoming) 2026

Short Fiction

C1. *The Mother of Gaia*. Bewildering Stories, Canada. 2025

FEATURED

F1. Honors student of the Department of Literature at UCSD. [Webpage] 2016

REFERENCES**John A. Stankovic**

B. P. America Professor

Department of Computer Science

University of Virginia

85 Engineer's Way, Charlottesville, VA 22903

434-982-2275

jas9f@virginia.edu

Jacobo Myerston

Associate Professor

Department of Literature

University of California, San Diego

UC San Diego Arts and Humanities Building, San Diego, CA 92161

jmyerston@ucsd.edu

Page duBois

Distinguished Professor

Department of Literature

University of California, San Diego

UC San Diego Arts and Humanities Building, San Diego, CA 92161

858-534-3379

pdubois@ucsd.edu